Society of St. Francis Xavier, Pilar's
Fr. Conceicao Rodrigues College of Engineering
Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai – 400 050
(Autonomous College affiliated to University of Mumbai)

# TE Comp A

## CSS TH ISE: Virtual Lab Development

All students must note that this is a part of theory ISE of 20 marks so all have to complete the below assigned work on or before 20/10/2026.The evaluation of this is schedule from 26/10/2026 to 31/10/2026 during the lab hours.

All students have to decide the group members from the same batch and provide the details in the below given link on or before 16/10/2026.

https://docs.google.com/document/d/1ZM5dJB4U0zB-OIbzb_683cqGgdjqeP33UVr_Ph96Jtc/edit?usp=sharing

## Guidelines:

1. Students shall work in groups of 5 members each.
2. All Five members of the group are expected to contribute meaningfully to the development of the virtual lab.
3. Experiments will be allotted to groups by the faculty.
4. Each group shall develop the complete virtual lab for the experiment assigned to them.
5. All groups shall use the same front-end software/tool for developing the virtual labs.
6. A common user interface, layout, colour scheme, and theme shall be followed by all groups to maintain uniformity across the virtual laboratory.
7. Each virtual lab should, as applicable, include:
    a. Aim and objectives of the experiment
    b. Relevant theoretical concepts
    c. Step-by-step procedure
    d. Interactive simulation/implementation
    e. Assessment/quiz
    f. References
    g. Feedback
8. The interface should be simple, intuitive, interactive, and user-friendly.
9. Navigation and terminology should be consistent across all experiments.
10. The virtual lab should incorporate interactive elements and should not merely display static content.
11. Students should verify the correctness of the virtual experiment before submission.

12. Groups shall demonstrate their virtual lab to the faculty as per the schedule communicated by the faculty.
13. Each group shall submit the required documentation along with the virtual lab.

## Rubrics for evaluation:

| Evaluation Criterion | 4- Excellent | 3- Good | 2- Satisfactory | 1- Needs Improvement |
|---|---|---|---|---|
| **1. Experiment Content & Accuracy** | Content is complete, accurate, well-structured, and covers objective, theory, procedure, observations, and results. No significant errors. | Content is mostly complete and accurate with minor omissions/errors. | Basic content is provided but has noticeable gaps or minor inaccuracies. | Content is incomplete, poorly organized, or contains significant errors. |
| **2. Functionality & Working** | Virtual lab works flawlessly; all inputs, processes, outputs, controls, and experiment steps function correctly. | Virtual lab works correctly with minor functional issues that do not significantly affect the experiment. | Basic functionality is available, but some features or experimental steps do not work correctly. | Major functionality is missing or the virtual lab does not work as intended. |
| **3. Interactivity & Simulation** | Highly interactive and provides realistic simulation; user inputs produce appropriate dynamic responses and meaningful feedback/visualization. | Good level of interaction and simulation with appropriate responses to most user inputs. | Limited interaction; simulation represents only basic aspects of the experiment. | Minimal/no interaction; largely static content with little or no meaningful simulation. |
| **4. Presentation & Demonstration** | Demonstration is clear, systematic, confident, and technically sound. Students explain the working effectively and answer questions accurately. | Demonstration is clear and well organized; students explain most aspects correctly and answer most questions. | Demonstration covers basic working but lacks clarity or depth; answers to questions are partially correct. | Demonstration is unclear/incomplete; students have limited understanding and are unable to answer questions adequately. |
| **5. Individual Contribution** | Student demonstrates excellent understanding of their assigned work, makes significant contribution, and independently explains and handles technical issues. | Student has good understanding and contribution; can explain most of their assigned work. | Student has limited understanding or contribution; requires assistance to explain their work. | Student demonstrates little/no understanding or contribution to the project. |

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## Allotment of Experiments:

| Sr. No. | Roll No. | Name of the Students | Title of the Experiment |
|---|---|---|---|
| 1 | | | 1. **To demonstrate the packet sniffing** attack. Simulated Wireshark-like interface<br><br>Where user can<br><br>• Capture packets<br>• Filter packets<br>• Examine headers<br>• Identify source/destination IP<br>• Identify protocols<br>• Analyze TCP/UDP packets<br>• Observe plaintext versus encrypted traffic |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |
| 6 | | | **2. Demonstrate ARP Spoofing,**how incorrect ARP mappings affect communication. |
| 7 | | | |
| 8 | | | |
| 9 | | | |
| 10 | | | |
| 11 | | | 3. Demonstrate the SQL Injection |
| 12 | | | |
| 13 | | | |
| 14 | | | |
| 15 | | | 4. Impact of vulnerable authentication |
| 16 | | | |
| 17 | | | |
| 18 | | | |
| 19 | | | Demonstrate the brute force and Frequency analysis attack/Breaking of any mono alphabetic cipher. |
| 20 | | | |
| 21 | | | |
| 22 | | | |
| 23 | | | |
| 24 | | | Demonstrate the Reading arbitrary files via path traversal |
| 25 | | | |
| 26 | | | |
| 27 | | | |

| | | | |
|---|---|---|---|
| 28 | | | Demonstrate the Buffer overflow attack |
| 29 | | | |
| 30 | | | |
| 31 | | | |
| 32 | | | Demonstrate the DoS and DDoS attack |
| 33 | | | |
| 34 | | | |
| 35 | | | |
| 36 | | | |
| 37 | | | Demonstrate the cross site scripting attack |
| 38 | | | |
| 39 | | | |
| 40 | | | |
| 41 | | | |
| 42 | | | Demonstrate the vulnerability of the system exploited by the Trojan horse. |
| 43 | | | |
| 44 | | | |
| 45 | | | |
| 46 | | | |
| 47 | | | Demonstrate the ICMP flood attack |
| 48 | | | |
| 49 | | | |
| 50 | | | |
| 51 | | | |
| 52 | | | Demonstrate Malicious Program Detection using Virus Signature and Behavioral Analysis |
| 53 | | | |
| 54 | | | |
| 55 | | | |
| 56 | | | |
| 57 | | | Demonstrate the effect of a SYN Flood attack on network availability. |
| 58 | | | |
| 59 | | | |
| 60 | | | |
| 61 | | | Demonstrate the known cipher text attack on RSA |
| 62 | | | |
| 63 | | | |
| 64 | | | |

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## Compilation Team:

| | |
|---|---|
| 1. | DSILVA LEHEL VIJIL |
| 2. | CABRAL ZIEL NOEL |
| 3. | KHARCHE ARJUN VIVEK |
| 4. | LOBO SANIKA OSWALD |
| 5. | RAUT ASHMAN CHETAN |
| 6. | PATADE ARNAV ANAND |
| 7. | THOMAS JUSTIN ROY |

## Common Template for each team:

```
experiment-template/
│
├── index.html
├── style.css
├── script.js
└── README.md
```

Each group shall develop its assigned cryptography experiment as an independent, self-contained web module using the common technology stack, UI template, folder structure, and input/output specifications provided by the integration team. The module must not modify the main application or other experiment modules. Each group shall submit the complete source code, experiment content, test cases, expected outputs, evaluation questions, quiz questions, and an integration README. The final integration team will combine the independently developed modules into a single Virtual Cryptography Laboratory application.

## Group: SQL Injection :

**Experiment ID:** EXP03

**Experiment Name:** SQL Injection Attack and Prevention

**Folder:**

/experiments/sql-injection/

Entry File:

index.html

Navigation Title:

SQL Injection Attack

## Group: SQL Injection

**Experiment ID:** EXP08

**Experiment Name:** SQL Injection Attack and Prevention

**Folder:**

/experiments/sql-injection/

**Entry File:**

index.html

**Navigation Title:**

SQL Injection Attack

**Short Description:**

Demonstrate SQL injection vulnerabilities using a simulated login/database environment and analyze techniques for preventing unauthorized database access.

**Required Libraries:**

None

**Input:**

Username
Password
SQL Query/Input

**Output:**

Authentication Result
Generated SQL Query

Database Response
Vulnerability Status

**Expected Navigation Link:**

/experiments/sql-injection/

**Learning Outcome:**

Understand how improper input validation can make applications vulnerable to SQL injection and demonstrate basic techniques for preventing the vulnerability.

**Virtual Lab Components:**

1. Login/Input Interface
2. Simulated Database
3. SQL Query Visualization
4. Attack Simulation
5. Vulnerability Detection
6. Prevention/Parameterized Query Demonstration
7. Result and Activity Log

**Use GitHub from the beginning**

Students submit their completed module through their **branch/pull request**.

**Use below color theme:**

Demonstrate SQL injection vulnerabilities using a simulated login/database environment and analyze techniques for preventing unauthorized database access.

**Required Libraries:**

None

**Input:**

Username
Password
SQL Query/Input

**Output:**

Authentication Result
Generated SQL Query
Database Response
Vulnerability Status

**Expected Navigation Link:**

/experiments/sql-injection/

**Learning Outcome:**

Understand how improper input validation can make applications vulnerable to SQL injection and demonstrate basic techniques for preventing the vulnerability.

**Virtual Lab Components:**

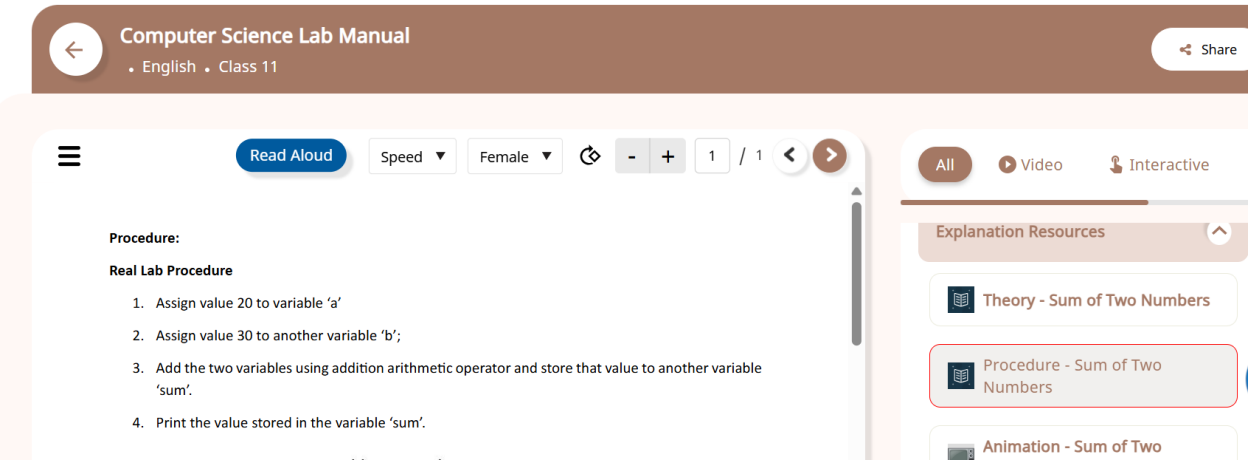
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# Common Template for Integration Team

```
crypto-virtual-lab/
│
├── index.html
│
├── css/
│   └── style.css
│
├── js/
│   └── common.js
│
├── experiments/
│
│   ├── Exp- 1/
│   │   ├── index.html
│   │   ├── script.js
│   │   └── README.md
│   │
│   ├── Exp-2/
│   │   ├── index.html
│   │   ├── script.js
│   │   └── README.md
│   │
│   ├── Exp-3/
│   │   ├── index.html
```

```
│   │   ├── script.js
│   │   └── README.md
│   │
│   └── Exp-4/
│       ├── index.html
│       ├── script.js
│       └── README.md
│
└── assets/
```

# Integration team responsibility

### Integration team WILL develop:

- Home page
- Experiment listing
- Navigation
- Common header/footer
- Common CSS/theme
- Experiment cards
- Routing/navigation
- About Virtual Lab
- Instructions
- Common quiz/evaluation framework, if applicable
- Linking all experiment modules
- Final testing

### Integration team WILL NOT develop any of the above experiments but they must provide the output like IIT Lab or Port Swigger lab

**Create one repository:**

```
cryptography-virtual-lab
```

Give each group a separate branch:

```
main
├── group-EXP-1
├── group- EXP-2
├── group- EXP-3
```

```
├── group- EXP-4
├── group- EXP-5

├── group- EXP-6
├── group- EXP-7
```

**The integration team works mainly on:**

The Integration Team is responsible for integrating all individual experiments developed by different student teams into a single, unified Virtual Lab platform. The team ensures that all experiments follow a common structure, design, navigation, and functionality.

The Integration Team works mainly on:

1. **Integration of Experiments**
    i. Collect and integrate all completed experiment modules into the main Virtual Lab.
    ii. Ensure that every experiment is accessible through the central navigation/menu.
    iii. Maintain the correct experiment IDs, folder structure, and entry files.

2. **Common User Interface**
    i. Maintain a consistent layout, navigation bar, buttons, fonts, icons, and color theme across all experiments.
    ii. Ensure that individual experiments look like part of the same Virtual Lab.

3. **Navigation Management**
    i. Create and maintain the main experiment listing/index page.
    ii. Add links to all experiments using the specified navigation titles and paths.
    iii. Verify that there are no broken links or incorrect paths.

4. **README and Documentation Verification**
    i. Collect the Integration README prepared by each experiment team.
    ii. Verify Experiment ID, Experiment Name, Folder, Entry File, Navigation Title, Description, Input, Output, and Required Libraries.
    iii. Maintain an overall integration document/README for the complete Virtual Lab.

5. **GitHub Repository Management**

i. Maintain the main GitHub repository and integration branch.
ii. Review student branches and pull requests.
iii. Merge completed and verified experiment modules into the main branch.
iv. Resolve merge conflicts and maintain proper version control.

6. **Compatibility Testing**

i. Check whether all experiments work correctly after integration.
ii. Test links, buttons, simulations, animations, forms, quizzes, and output displays.
iii. Ensure that experiments do not interfere with each other.

7. **Responsive Design Testing**

i. Verify that the integrated Virtual Lab works properly on different screen sizes.
ii. Check compatibility with commonly used browsers such as Chrome, Edge, and Firefox.

8. **Final Quality Assurance**

i. Identify missing files, broken functionality, inconsistent UI, and integration errors.
ii. Coordinate with respective experiment teams to correct issues.
iii. Perform final testing before deployment.

9. **Deployment Preparation**

i. Organize the final project folder structure.
ii. Prepare the repository for hosting/deployment.
iii. Verify that all experiments are available through their expected navigation links.

All team members are expected to divide the work in the team and see the smooth implementation of the virtual lab.

**Subject Incharge**

**Dr.Smita Ambarkar**